

```
In [38]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn import linear_model
from sklearn.metrics import confusion_matrix, accuracy_score, recall_score
```

```
In [14]: df = pd.read_csv("/home/student/Downloads/Social_Network_Ads.csv")
```

```
In [15]: df
```

```
Out[15]:
```

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
...	...	...	...	...	...
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

400 rows × 5 columns

```
In [16]: df.shape
```

```
Out[16]: (400, 5)
```

```
In [17]: df.describe()
```

```
Out[17]:
```

	User ID	Age	EstimatedSalary	Purchased
count	4.000000e+02	400.000000	400.000000	400.000000
mean	1.569154e+07	37.655000	69742.500000	0.357500
std	7.165832e+04	10.482877	34096.960282	0.479864
min	1.556669e+07	18.000000	15000.000000	0.000000
25%	1.562676e+07	29.750000	43000.000000	0.000000
50%	1.569434e+07	37.000000	70000.000000	0.000000
75%	1.575036e+07	46.000000	88000.000000	1.000000
max	1.581524e+07	60.000000	150000.000000	1.000000

```
In [28]: X = df[['Age', 'EstimatedSalary']]
y = df['Purchased']
```

```
In [29]: from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=0
)
```

```
In [52]: from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)
```

```
In [53]: from sklearn.linear_model import LogisticRegression
classifier = LogisticRegression(random_state=0)
classifier.fit(X_train, y_train)
```

```
Out[53]: ▼      LogisticRegression
LogisticRegression(random_state=0)
```

```
In [46]: y_pred = classifier.predict(X_test)
```

```
In [47]: from sklearn.metrics import confusion_matrix
cm = confusion_matrix(y_test, y_pred)

print("Confusion Matrix:\n", cm)
```

Confusion Matrix:

```
[[65  3]
 [ 8 24]]
```

```
In [48]: TN = cm[0][0]
FP = cm[0][1]
FN = cm[1][0]
TP = cm[1][1]

print("\nTP:", TP)
print("TN:", TN)
print("FP:", FP)
print("FN:", FN)
```

TP: 24
TN: 65
FP: 3
FN: 8

```
In [49]: accuracy = (TP + TN) / (TP + TN + FP + FN)
error_rate = (FP + FN) / (TP + TN + FP + FN)
precision = TP / (TP + FP)
recall = TP / (TP + FN)
print("Precision:", precision)
print("Recall:", recall)
print("Accuracy:", accuracy)
print("Error_Rate:", error_rate)
```

Precision: 0.8888888888888888
Recall: 0.75
Accuracy: 0.89
Error_Rate: 0.11

```
In [51]: from sklearn.metrics import classification_report
print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

Classification Report:					
	precision	recall	f1-score	support	
	0	0.89	0.96	0.92	68
	1	0.89	0.75	0.81	32
accuracy				0.89	100
macro avg				0.89	100
weighted avg				0.89	100